

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/website_traffic_2025.csv")
```

```
df.head()
```

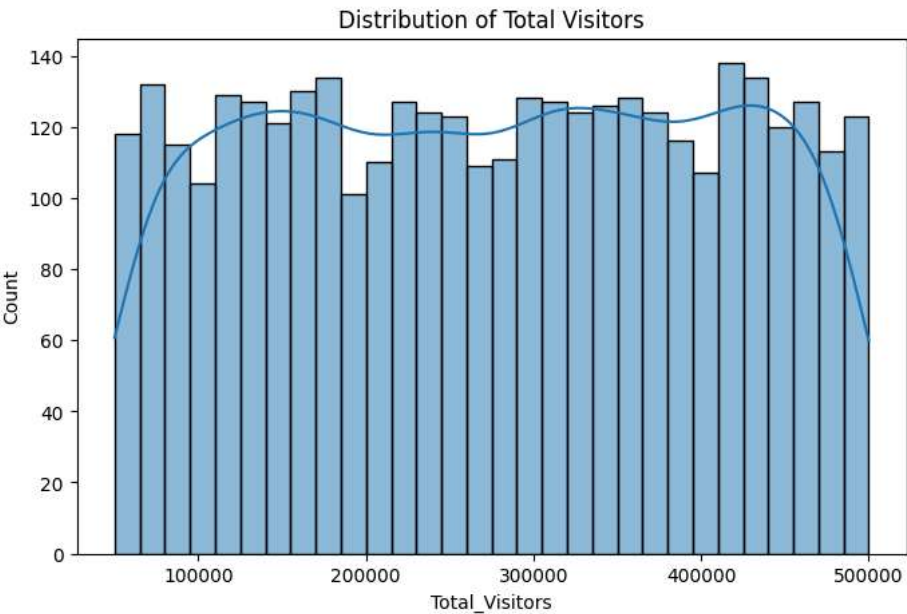
	Date	Website_Name	Total_Visitors	Unique_Visitors	Page_Views	Bounce_Rate	Avg_Time_Spent	Traffic_Source	Device_Type	Co
0	2025-01-01	Amazon	77453	48523	154906	63.45	6.26	Referral	Tablet	Ge
1	2025-01-01	YouTube	224167	134335	1793336	27.60	4.16	Social Media	Mobile	C
2	2025-01-01	Netflix	455582	205199	2733492	43.88	2.11	Google	Mobile	Au

```
#Univariate Analysis
df['Total_Visitors'].describe()
```

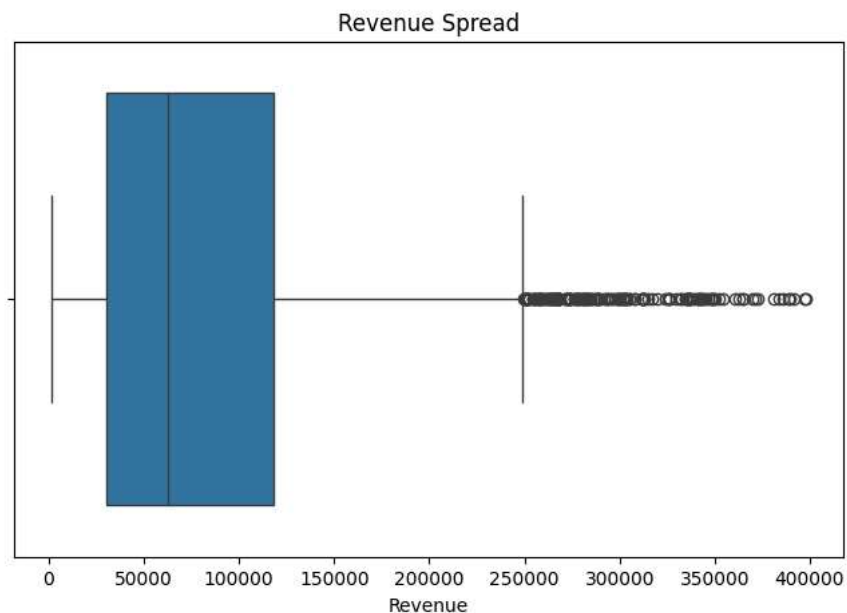
Total_Visitors	
count	3650.000000
mean	276104.724384
std	129928.231713
min	50182.000000
25%	164027.250000
50%	278116.500000
75%	389534.500000
max	499985.000000

dtype: float64

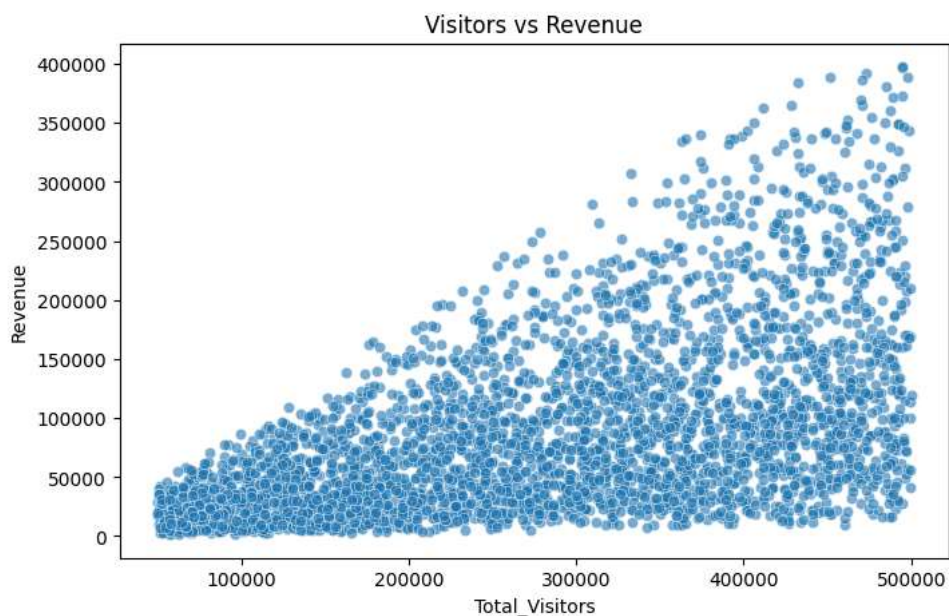
```
plt.figure(figsize=(8,5))
sns.histplot(df['Total_Visitors'], bins=30, kde=True)
plt.title("Distribution of Total Visitors")
plt.show()
```



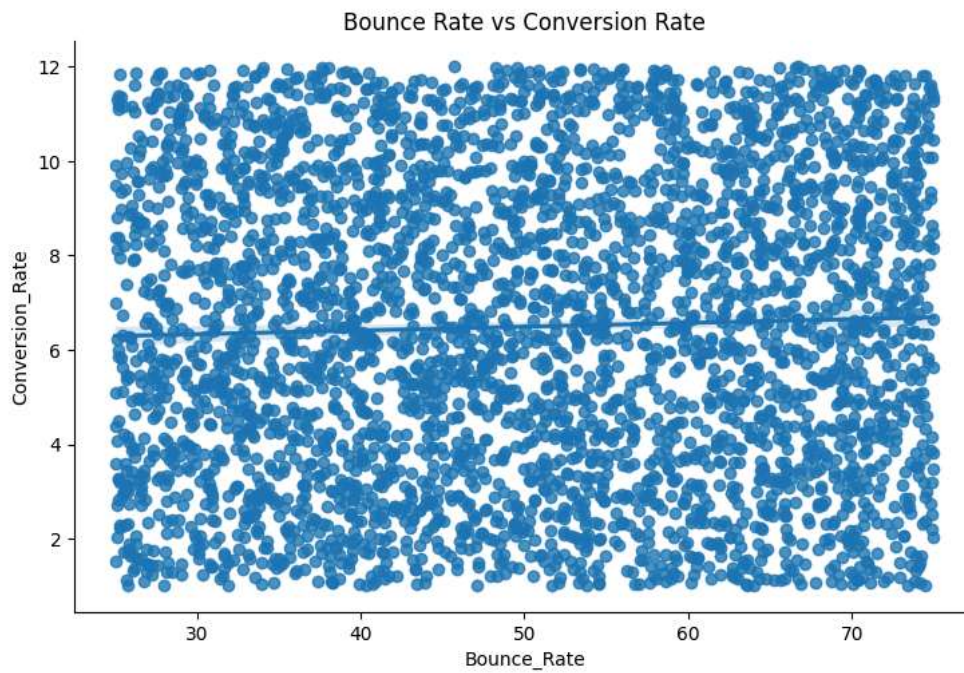
```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Revenue'])
plt.title("Revenue Spread")
plt.show()
```



```
#Bivariate Analysis
plt.figure(figsize=(8,5))
sns.scatterplot(x='Total_Visitors', y='Revenue', data=df, alpha=0.6)
plt.title("Visitors vs Revenue")
plt.show()
```



```
sns.lmplot(x='Bounce_Rate', y='Conversion_Rate', data=df, aspect=1.5)
plt.title("Bounce Rate vs Conversion Rate")
plt.show()
```

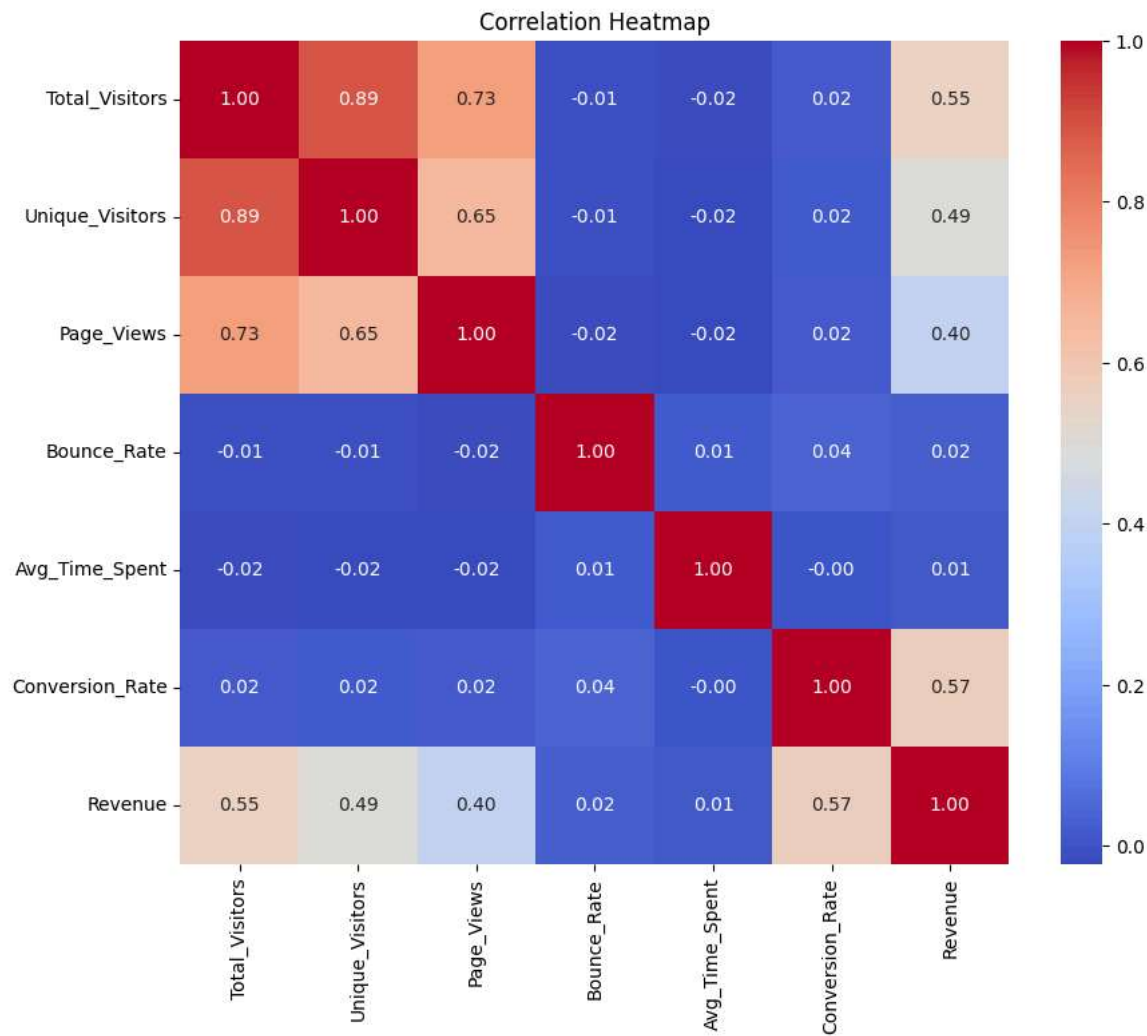


```
df.groupby('Device_Type')['Avg_Time_Spent'].mean()
```

	Avg_Time_Spent
Device_Type	
Desktop	10.212454
Mobile	9.980562
Tablet	10.080251

dtype: float64

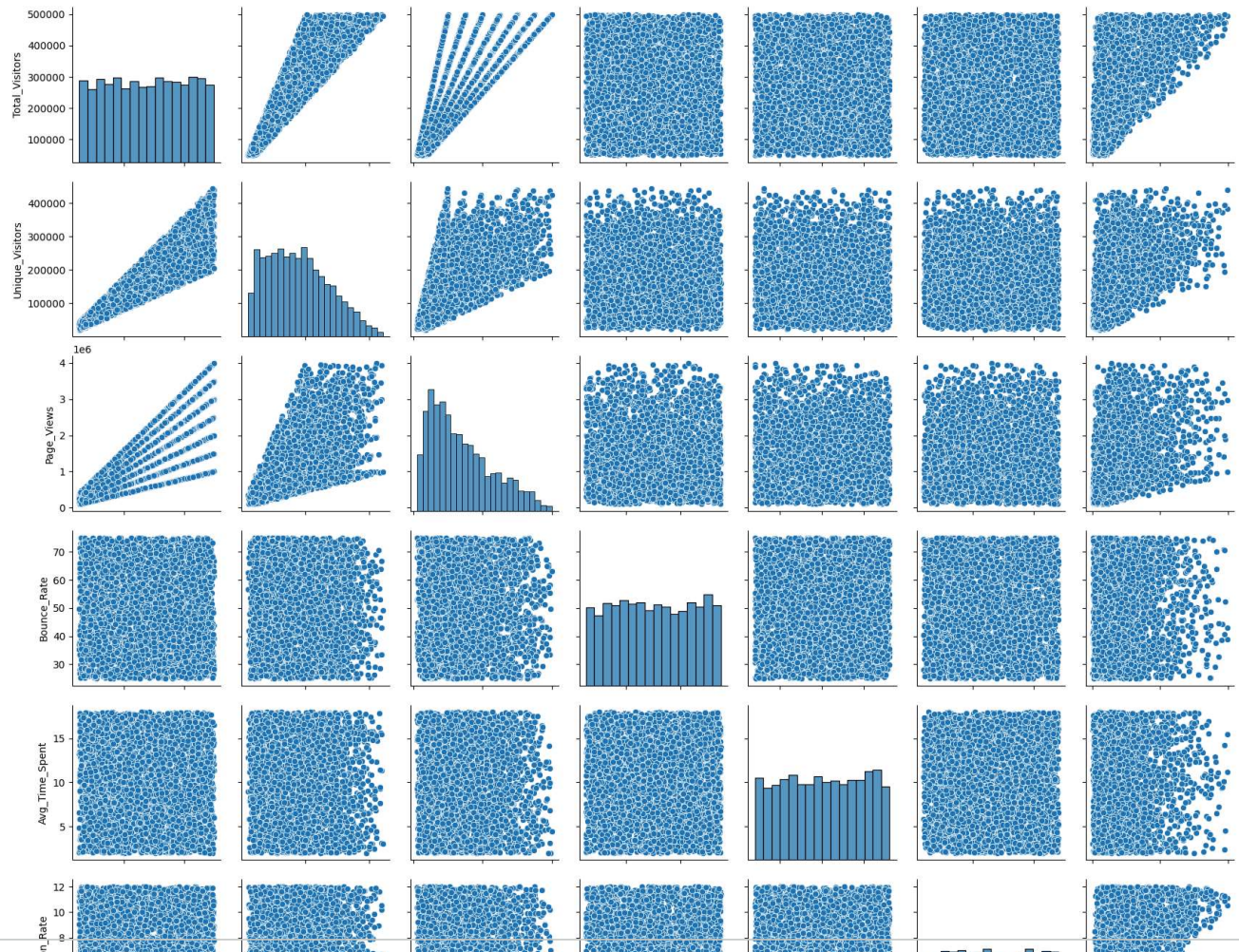
```
#Multivariate Analysis
# Correlation heatmap
plt.figure(figsize=(10,8))
sns.heatmap(df.select_dtypes(include='number').corr(),
            annot=True,
            cmap="coolwarm",
            fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



```
#Pairplot
sns.pairplot(df)
```



```
<seaborn.axisgrid.PairGrid at 0x7b93562a2300>
```



```
sns.scatterplot(x='Total_Visitors',
                y='Revenue',
                hue='Conversion_Rate',
                data=df)
```

